Chen 2020 NPC — analysis summary

RUN 5 OF 7

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony on sample (θ=4, n_pcs=19): iLISI 1.08 → 2.33 of a 3.23 ceiling (72%; ceiling <8 because P11+P01=72% of cells). Single connected kNN component over all 3,674 cells — continuous, sample-integrated. No patient islands.

Gate 2 — module geography → GO (non-branching)

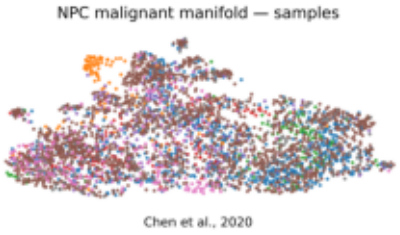
Gast/pSom overlap, Crest inverse terminal gradient. Per-node-cluster corr: Gast~pSom = +0.13 (OVERLAP, not cross-repression); Gast~Crest = −0.82; pSom~Crest = −0.50. Curve pseudotime: corr(t,Gast)=−0.58, corr(t,pSom)=−0.11, corr(t,Crest)=+0.44 — a single Gast/pSom-high → Crest-high axis. Cycling confound cleared (Kürten lesson): Gast~G2M = −0.49, progenitor is quiescent.

Gate 3 — hub cross-repression → NA

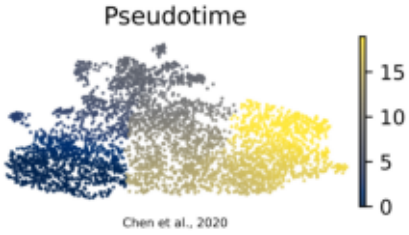
No bifurcation; branching-only correlation / modules notebooks not produced (spec §6).

STEP FIGURES

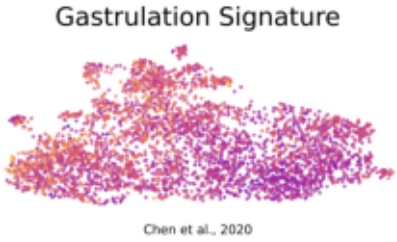
Manifold — samples



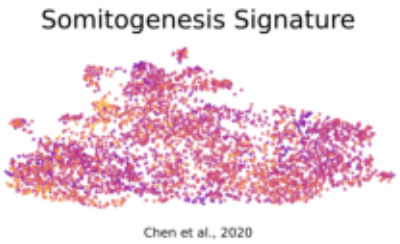
Pseudotime



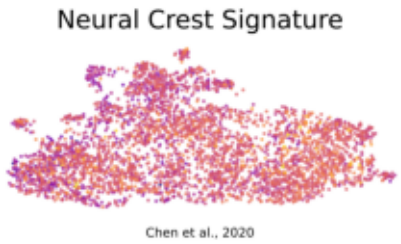
Gastrulation



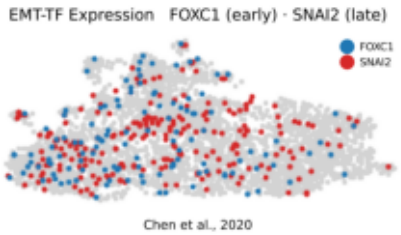
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Nasopharyngeal carcinoma (Chen et al., 2020), 10x — 14 samples; 3,674 malignant cells retained (of 3,763; 8 samples ≥107 cells).

Verdict.
The embryonic pattern reproduces as the NON-BRANCHING case: a single developmental axis from a Gast/pSom-high progenitor to a Crest-high terminus (structurally the Kürten 2021 outcome, not Puram's branched hub-and-spoke). Gate 1 GO / Gate 2 GO / Gate 3 NA. Detectable EMT-TFs FOXC1, SNAI2, PRRX1, ZEB2 (canonical SNAI1/ZEB1/TWIST below detection).

Parameter choices

Min-cell threshold	107 (count-gap rule, 8/14 samples)	**Scaling**	no sc.pp.scale (log1p, reference-matched)
Genes	1,290 overdispersed	**Embedding**	PCA n_pcs=19; Harmony θ=4
Trajectory	scFates curve Nodes=10; root = Gast/pSom-high tip	**Signatures**	score_genes, union control pool
EMT-TFs	FOXC1 / SNAI2 / PRRX1 / ZEB2 (canonical SNAI1/ZEB1/TWIST below floor)		